BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 6477 / 6474

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6477

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The dimension of ρgh is similar as that of:

(A) Power
(B) Torque
(C) Pressure
(D) Force

Q2. The wavelength of wave produced by microwave oven is:

(A) 6cm
(B) 12cm
(C) 24cm
(D) 50cm

Q3. Speed of sound in air at S.T.P. is:

(A) 280 m/s
(B) 330 m/s
(C) 332 m/s
(D) 350 m/s

Q4. Half wavelength corresponds to:

(A) 0°
(B) 90°
(C) 180°
(D) 360°

Q5. Which cannot be polarized?

(A) Sound waves
(B) X-rays
(C) Light waves
(D) Radio waves

Q6. The first person who attempted to measure the speed of light was:

(A) Newton
(B) Galileo
(C) Huygen
(D) Michelson

Q7. Boltzman constant "K" has the same unit as:

(A) Pressure
(B) Energy
(C) Temperature
(D) Entropy

Q8. If temperature of the sink decreases, then efficiency of Carnot engine:

(A) Increases
(B) Decreases
(C) Remains the same
(D) First increases then decreases

Q9. Which is not a base unit in SI units?

(A) Ampere
(B) Joule
(C) Kilogram
(D) Kelvin

Q10. If error in measurement of radius of circle is 2%, then permissible error in its area will be:

(A) 1%
(B) 2%
(C) 3%
(D) 4%

Q11. If Ax = Ay, then angle between $\bar{A}$ and x-axis is:

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q12. In which quadrant vector -2î -3ĵ lies?

(A) 1st
(B) 2nd
(C) 3rd
(D) 4th

Q13. Impulse has same unit as that of:

(A) Mass
(B) Energy
(C) Force
(D) Linear momentum

Q14. The range of projectile is same for:

(A) 10°, 70°
(B) 20°, 50°
(C) 25°, 65°
(D) 30°, 70°

Q15. Which one is non-renewable source of energy?

(A) Tides
(B) Biomass
(C) Waves
(D) Oil

Q16. Rotational K.E. of disc is given by:

(A) $\frac{1}{2}mv^2$
(B) $\frac{1}{4}mv^2$
(C) $\sqrt{gh}$
(D) $\sqrt{(4/3\ gh)}$

Q17. If a body of mass 10kg is falling freely, its apparent weight will be:

(A) Zero
(B) 10N
(C) 98N
(D) 980N

Physics (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1111-XI124

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Why do we find it useful to have two units for the amount of substance, the kilogram and the mole?

(ii) Does a dimensional analysis give any information on constant of proportionality that may appear in an algebraic expression? Explain.

(iii) Write the dimensions of (a) pressure (b) density.

(iv) If percentage uncertainty in radius of sphere is 0.4%, then what will be total uncertainty in its volume?

(v) Can a body rotate about its center of gravity under the action of its weight?

(vi) Name three conditions that could make, $A_1 \times A_2 = 0$

(vii) Draw the diagram of two cases in which components of a vector are equal in magnitude.

(viii) Explain the circumstances in which the velocity $\vec{v}$ and acceleration $\vec{a}$ of a car are (a) $\vec{v}$ is zero but $\vec{a}$ is not zero. (b) $\vec{a}$ is zero but $\vec{v}$ is not zero.

(ix) At what point or points in its path does a projectile have its minimum speed, its maximum speed?

(x) Which quantities are assumed to be constant in projectile motion?

(xi) What sort of energy is in (a) compressed spring (b) water in a high dam?

(xii) A girl drops a cup from a certain height, which breaks into pieces. What energy changes are involved?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Explain how many minimum number of geostationary satellites are required for global coverage of TV transmission.

(ii) Satellites orbiting at different altitudes have different time periods. Explain why?

(iii) Why is it difficult for a car to turn round a corner at high speed than at lower speed?

(iv) A 1000kg car moves with a speed of $40ms^{-1}$ round a curve of radius 100m. Find the necessary centripetal force.

(v) Explain how the swing is produced in a fast moving cricket ball?

(vi) What are systolic and diastolic pressures? Also give values.

(vii) Under what conditions, does the addition of two simple harmonic motions produce a resultant, which is also simple harmonic?

(viii) What will be the frequency of a simple pendulum if its length is 1m at place where g = $9.8ms^{-2}$?

(ix) Explain briefly the example of electrical resonance.

(x) How beats are useful in tuning musical instruments?

(xi) Differentiate between red shift and blue shift.

(xii) How the frequency of a string of a musical instrument can be changed?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Can visible light produce interference fringes? Explain.

(ii) Why the polaroid sunglasses are better than ordinary sunglasses?

(iii) Differentiate between a ray and a wave front.

(iv) Why would it be advantageous to use blue light with a compound microscope?

(v) If a person was looking through a telescope at the full moon, how would the appearance of the moon be changed by covering half of the objective lens?

(vi) What are the necessary conditions for total internal reflection?

(vii) Why specific heat at constant pressure is greater than specific heat at constant volume?

(viii) Why does pressure of a gas in a car tyre increase when it is driven through some distance?

(ix) Explain adiabatic process with two examples.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.

GROUP – II

Physics (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6474

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Cp - Cv = :

(A) Plank's constant
(B) Molar gas constant
(C) General gas constant
(D) Boltzmann constant

Q2. Which remains constant in an adiabatic process:

(A) Volume
(B) Entropy
(C) Pressure
(D) Temperature

Q3. Least distance of distinct vision for normal eye is:

(A) 15 cm
(B) 125 cm
(C) 25 cm
(D) 25 m

Q4. Fringe spacing increases if we use:

(A) Green light
(B) Red light
(C) Yellow light
(D) Blue light

Q5. With increase of temperature sound speed:

(A) Remains constant
(B) Increases
(C) Becomes zero
(D) Decreases

Q6. Half wave length corresponds to:

(A) 0°
(B) 90°
(C) 180°
(D) 360°

Q7. The wave form of SHM is:

(A) A square wave
(B) Sine wave
(C) Cosine wave
(D) Tangent wave

Q8. SI units of viscosity are:

(A) $kg^{-1}\ ms^{-1}$
(B) $kg^{-1}m^{-1}s$
(C) $kg\ m^{-1}\ s^{-1}$
(D) $kg\ m\ s^{-1}$

Q9. Centripetal force performs:

(A) Minimum work
(B) Maximum work
(C) No work
(D) Negative work

Q10. Which quantity is dimension less?

(A) Centripetal force
(B) Angular velocity
(C) Angular displacement
(D) Angular acceleration

Q11. Which is non-conservative force?

(A) Electrical force
(B) Gravitational force
(C) Frictional force
(D) Magnetic force

Q12. SI unit of impulse is equivalent to that of:

(A) Force
(B) Velocity
(C) Momentum
(D) Acceleration

Q13. Which formula is true?

(A) $m = a/F$
(B) $F = m/a$
(C) $a = F/m$
(D) $a = m/F$

Q14. Magnitudes of cross product and dot product of two vectors are equal. The angle between the vectors is:

(A) 0°
(B) 45°
(C) 180°
(D) 60°

Q15. First condition of equilibrium implies that:

(A) $\sum F = 0$
(B) $\sum Fx = 0$
(C) $\sum Fy = 0$
(D) $\sum Fx = \sum Fy$

Q16. Significant figures in 8.70×10^{4} kg are:

(A) 5
(B) 4
(C) 3
(D) 2

Q17. A light year is the distance light travels in one year. How many meters are there in one light year?

(A) 9.5×10^{-15} m
(B) 9.5×10^{15} km
(C) 9.5×10^{15} cm
(D) 9.5×10^{15} m

Physics (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1112-XI124

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Write the dimensions of pressure and density.

(ii) Name several repetitive phenomenon occurring in nature which could serve as reasonable time standard.

(iii) How many meters are there in one light year? Explain.

(iv) What are the characteristics of ideal standard?

(v) The vector sum of three vectors gives a zero resultant. What can be orientation of the vectors?

(vi) Can a body rotate about its center of gravity under the action of its weight?

(vii) If $\vec{A} = 3\hat{i} - 5\hat{j}$, $\vec{B} = 7\hat{k}$, find $(\vec{A} \times \vec{B})$

(viii) Define impulse and show that how it is related to linear momentum?

(ix) Explain the circumstances in which the velocity $\vec{v}$ and acceleration $\vec{a}$ of a car are perpendicular to one another.

(x) What is the effect on the speed of a fighter plane chasing another when it opens the fire?

(xi) When a rocket re-enters the atmosphere, its nose cone becomes very hot. Where does this heat energy come from?

(xii) Prove that $P = \vec{F} \cdot \vec{v}$

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) What is the venturi relation? Which quantity is measured using this relation?

(ii) How does swing is produced in a tennis ball?

(iii) Two cylinders of equal mass but with different diameters, which has greater rotational inertia?

(iv) What do you know about GPS and its use?

(v) What is an orbital velocity? What does effect of mass of satellite on value of orbital velocity?

(vi) How do you find direction of angular momentum and angular velocity in simple situation?

(vii) Why does the oscillation of a vibrating body eventually stop?

(viii) If a pendulum vibrates with frequency ' f '. What does effect on its angular frequency, if its time period is doubled?

(ix) What does information is determined by phase of a vibrating body?

(x) Describe the term crest, trough, node and antinode.

(xi) How does the speed of distant stars and galaxies are calculated?

(xii) In the phenomenon of stationary waves, if string vibrates in more and more loops, what would you conclude about its frequency and wavelength?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) What conditions must be met to observe the interference of light?

(ii) Why the polaroid sunglasses are better than ordinary sunglasses?

(iii) Justify that a path difference $\lambda/4$ is neither associated with constructive interference nor destructive interference of light.

(iv) How the power is lost in optical fiber through dispersion? Explain.

(v) How the light propagates with in a flexible glass fiber?

(vi) Describe briefly how light is refracted in continuous refraction?

(vii) Can the mechanical energy be converted completely into heat energy? If so, give an example.

(viii) Calculate the change in internal energy when 42J heat energy is transferred to the system during the expansion and 32J work is done on the piston.

(ix) Does entropy of a system increase or decrease due to friction? Explain.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.